

- Q.27 What is state machine. (CO11)
 Q.28 Write application of A/D converter. (CO1)
 Q.29 Differentiate between Latch & FF. (CO7)
 Q.30 Explain any one type of Encoder. (CO6)
 Q.31 Explain 2 bit comparator. (CO5)
 Q.32 Differentiate between analog & digital sigma. (CO3)
 Q.33 What is 7-Segment display. (CO3)
 Q.34 Explain about IC 7495. (CO8)
 Q.35 What is programmable counter. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain with diagram the working of Dual – Slope A/D converter. (CO10)
 Q.37 Simplify using K-Map. (CO4)
 $F(A, B, C, D) = \sum m(1, 3, 5, 7, 11, 13) + d(4, 6)$
 Q.38 Explain with diagram any one type of Decade Counter. (CO9)

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Time : 3 Hrs.

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SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 In 16:1 mux, how many select lines are required (CO6)
 a) 3 b) 4
 c) 1 d) 2
 Q.2 JK FF consists of ___ inputs. (CO7)
 a) 2 b) 3
 c) 1 d) 4
 Q.3 The number of digits in hexadecimal system is (CO2)
 a) 8 b) 10
 c) 16 d) 2
 Q.4 Full adder circuit has ___ inputs. (CO5)
 a) 4 b) 3
 c) 2 d) 6

- Q.5 There are _____ number of cells in 4-Variable K-map. (CO4)
- a) 8 b) 4
- c) 12 d) 16
- Q.6 The NOR gate is OR gate followed by (CO7)
- a) NOT gate b) OR gate
- c) AND gate d) EX-Or gate
- Q.7 A FF stores _____ bit of information. (CO7)
- a) Two b) Four
- c) One d) Three
- Q.8 The Binary number 11001 is equivalent to decimal number. (CO2)
- a) 24 b) 27
- c) 26 d) 25
- Q.9 Which is the 1's complement of 101101. (CO2)
- a) 110110 b) 010010
- c) 100011 d) 011011
- Q.10 Which gate is called as universal gate. (CO2)
- a) NAND b) OR
- c) AND d) NDT

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Draw, truth table of SR FF? (CO7)
- Q.12 Full form of PIPO _____. (CO8)
- Q.13 State Demorgan's theorem. (CO3)
- Q.14 BCD code of $(24)_{10}$ is _____. (CO2)
- Q.15 Draw truth table of NAND gate. (CO3)
- Q.16 Write any two advantages of digital sign. (CO3)
- Q.17 Full form of TTL is _____. (CO3)
- Q.18 Full form of ASCII is _____. (CO2)
- Q.19 Draw truth table of Half adder. (CO5)
- Q.20 What is Parity.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain NOR gate as Universal gate. (CO3)
- Q.22 Convert $(64)_{10} = (?)_2 = (?)_8$ (CO2)
- Q.23 Explain D-Latch with diagram. (CO7)
- Q.24 Explain PIPO shift register. (CO8)
- Q.25 Explain error detection. (CO2)
- Q.26 Simplify
 $Y = ABC + \bar{A}BC + A\bar{B}C + AC.$ (Co3)